1. General measure theory

In this chapter we shall introduce some general measure-theoretic concepts, terminology, and results which will be needed later on. But we shall also assume that the reader is familiar with basic measure theory. Most of the material needed can be found in several standard books such as Halmos [1], Hewitt and Stromberg [1], Munroe [1], Royden [1], Rudin [1], and many others including Federer [3] and Rogers [1]. Many of the proofs will be omitted. In this chapter we shall also introduce a great deal of notation and terminology to be used throughout the book. We shall generally follow the most standardized terminology of measure theory with one notable exception. Following Federer and Rogers we shall call measure what is usually called outer measure.

Some basic notation

We shall work in a metric space X with a metric d, although most of the measure theory presented here goes through in more general settings. Later on we shall however mainly stay in the euclidean n-space $\mathbf{R}^n$. Here are the basic notations used in metric spaces throughout this book.

The closed and open balls with centre $x \in X$ and radius r, $0 < r < \infty$, are denoted by

$$\begin{aligned} B(x,r) &= \{y \in X : d(x,y) \leq r\}, \\ U(x,r) &= \{y \in X : d(x,y) < r\}. \end{aligned}$$

In $\mathbf{R}^n$ we also set

$$B(r) = B(0,r),\ U(r) = U(0,r),\ S(x,r) = \partial B(x,r) \text{ and } S(r) = S(0,r).$$

The diameter of a non-empty subset A of X is

$$d(A) = \sup\{d(x,y) : x, y \in A\}.$$

We agree $d(\emptyset) = 0$. If $x \in X$ and A and B are non-empty subsets of X, the distance from x to A and the distance between A and B are, respectively,

$$\begin{aligned} d(x,A) &= \inf\{d(x,y) : y \in A\}, \\ d(A,B) &= \inf\{d(x,y) : x \in A, y \in B\}. \end{aligned}$$

For $\varepsilon > 0$ the closed ε-neighbourhood of A is

$$A(\varepsilon) = \big\{x \in X : d(x,A) \leq \varepsilon\big\}.$$

Measures

A measure for us will be a non-negative, monotonic, subadditive set function vanishing for the empty set.

1.1. Definition. *A set function* μ: $\{A : A \subset X\} \to [0, \infty] = \{t : 0 \leq t \leq \infty\}$ *is called a measure if*

(1) $\mu(\emptyset) = 0$,
(2) $\mu(A) \leq \mu(B)$ *whenever* $A \subset B \subset X$,
(3) $\mu\Big(\bigcup\limits_{i=1}^{\infty} A_i\Big) \leq \sum\limits_{i=1}^{\infty} \mu(A_i)$ *whenever* $A_1, A_2, \cdots \subset X$.

Usually in measure theory a measure means a non-negative countably additive set function defined on some σ-algebra of subsets of X, which need not be the whole power set $\{A : A \subset X\}$. However, considering measures in the sense of Definition 1.1 is a convenience rather than a restriction. That is, if ν is a countably additive non-negative set function on a σ-algebra $\mathcal{A}$ of subsets of X, it can be extended to a measure ν^* on X (in the sense of Definition 1.1) by

$$\nu^*(A) = \inf\{\nu(B) : A \subset B \in \mathcal{A}\}, \tag{1.2}$$

see Exercise 1. On the other hand, a measure μ gives a countably additive set function when restricted to the σ-algebra of μ measurable sets.

1.3. Definition. *A set* $A \subset X$ *is* μ *measurable if*

$$\mu(E) = \mu(E \cap A) + \mu(E \setminus A) \quad \text{for all } E \subset X.$$

We collect the well-known basic properties of measurable sets in the following theorem.

1.4. Theorem. *Let* μ *be a measure on* X *and let* $\mathcal{M}$ *be the family of all* μ *measurable subsets of* X.

(1) $\mathcal{M}$ *is a* σ*-algebra, that is,*

(i) $\emptyset \in \mathcal{M}$ *and* $X \in \mathcal{M}$,
(ii) *if* $A \in \mathcal{M}$, *then* $X \setminus A \in \mathcal{M}$,
(iii) *if* $A_1, A_2, \cdots \in \mathcal{M}$, *then* $\bigcup_{i=1}^{\infty} A_i \in \mathcal{M}$.

(2) *If* $\mu(A) = 0$, *then* $A \in \mathcal{M}$.
(3) *If* $A_1, A_2, \dots \in \mathcal{M}$ *are pairwise disjoint, then*

$$\mu\Big(\bigcup_{i=1}^{\infty} A_i\Big) = \sum_{i=1}^{\infty} \mu(A_i).$$

(4) *If* $A_1, A_2, \dots \in \mathcal{M}$, *then*

(i) $\mu\Big(\bigcup_{i=1}^{\infty} A_i\Big) = \lim_{i\to\infty} \mu(A_i)$ *provided* $A_1 \subset A_2 \subset \dots$,

(ii) $\mu\Big(\bigcap_{i=1}^{\infty} A_i\Big) = \lim_{i\to\infty} \mu(A_i)$ *provided* $A_1 \supset A_2 \supset \dots$ *and* $\mu(A_1) < \infty$.

It is also good to remember that the first statement of (4) holds without the measurability assumption if μ is *regular*, that is, for every $A \subset X$ there is a μ measurable set $B \subset X$ such that $A \subset B$ and $\mu(A) = \mu(B)$.

Recall that the family of *Borel sets* in X is the smallest σ-algebra containing the open (or equivalently closed) subsets of X. We shall often consider measures with some of the following properties.

1.5. Definition. *Let* μ *be a measure on* X.

(1) μ *is locally finite if for every* $x \in X$ *there is* $r > 0$ *such that* $\mu(B(x, r)) < \infty$.
(2) μ *is a Borel measure if all Borel sets are* μ *measurable.*
(3) μ *is Borel regular if it is a Borel measure and if for every* $A \subset X$ *there is a Borel set* $B \subset X$ *such that* $A \subset B$ *and* $\mu(A) = \mu(B)$.
(4) μ *is a Radon measure if it is a Borel measure and*

(i) $\mu(K) < \infty$ *for compact sets* $K \subset X$,
(ii) $\mu(V) = \sup\{\mu(K) : K \subset V \text{ is compact}\}$ *for open sets* $V \subset X$,
(iii) $\mu(A) = \inf\{\mu(V) : A \subset V,\ V \text{ is open}\}$ *for* $A \subset X$.

We shall give a few simple examples. Many others will be encountered later on.

1.6. Examples.

(1) The Lebesgue measure $\mathcal{L}^n$ on $\mathbf{R}^n$ is a Radon measure.

(2) The Dirac measure δ_a at a point $a \in X$ is defined by $\delta_a(A) = 1$, if $a \in A$, $\delta_a(A) = 0$, if $a \notin A$ (that is, $\delta_a(A) = \chi_A(a)$). It is a Radon measure on any metric space X.
(3) The counting measure n on X is defined by letting $n(A)$ be the number of elements in A, possibly ∞. It is Borel regular on any metric space X, but it is a Radon measure only if every compact subset of X is finite, that is, X is discrete.

In general, Radon measures are always Borel regular as a rather immediate consequence of the definition. The converse is not true as the above example (3) shows. However, locally finite Borel regular measures in complete separable metric spaces are Radon measures, see e.g. Jacobs [1, Theorem V.5.3] or Schwartz [1, Part I, § II.3]. In $\mathbf{R}^n$ this will be stated in Corollary 1.11. Clearly in $\mathbf{R}^n$ the local finiteness means that compact sets have finite measure.

Borel measures in metric spaces are often called metric (outer) measures, because the following, Carathéodory's, criterion gives a very convenient necessary and sufficient metric condition for the measurability of Borel sets.

1.7. Theorem. *Let μ be a measure on X. Then μ is a Borel measure if and only if*

$$\mu(A \cup B) = \mu(A) + \mu(B) \quad \textit{whenever } A, B \subset X \textit{ with } d(A,B) > 0.$$

The proof of the more essential "if" part is given in many text-books, e.g. Munroe [1], Falconer [4], Federer [3], L. Simon [1]. The easier "only if" part is left as an exercise.

Given a measure μ and a subset A of X we can form a new measure by restricting μ to A.

1.8. Definition. *The restriction of a measure μ to a set $A \subset X$, $\mu \llcorner A$, is defined by*

$$(\mu \llcorner A)(B) = \mu(A \cap B) \quad \textit{for } B \subset X.$$

It is clear that $\mu \llcorner A$ is a measure. Many of the relations between μ and $\mu \llcorner A$ are easy to derive. For example,

1.9. Theorem.

(1) *Every μ measurable set is also $\mu \llcorner A$ measurable.*
(2) *If μ is Borel regular and A is μ measurable with $\mu(A) < \infty$, then $\mu \llcorner A$ is Borel regular.*

Proof. The first statement is readily checked from the definitions. Note that A can be quite arbitrary there. We prove the second part.

Let B be a Borel set with $A \subset B$ and $\mu(A) = \mu(B)$. Then $\mu(B \setminus A) = 0$. Given $C \subset X$ let D be a Borel set with $B \cap C \subset D$ and $\mu(B \cap C) = \mu(D)$. Then $C \subset D \cup (X \setminus B) = E$, say, and

$$\begin{aligned}(\mu \llcorner A)(E) &\leq \mu(B \cap E) = \mu(B \cap D) \leq \mu(D)\\ &= \mu(B \cap C) = \mu(A \cap C) = (\mu \llcorner A)(C).\end{aligned}$$

Thus $(\mu \llcorner A)(E) = (\mu \llcorner A)(C)$, and so $\mu \llcorner A$ is Borel regular. □

The following approximation theorem will be extremely useful, see e.g. Evans and Gariepy [1, Theorem 1.1.4], Federer [3, Theorem 2.2.2] or L. Simon [1, Theorem 1.3].

1.10. Theorem. *Let μ be a Borel regular measure on X, A a μ measurable set, and $\varepsilon > 0$.*

(1) *If $\mu(A) < \infty$, there is a closed set $C \subset A$ such that $\mu(A \setminus C) < \varepsilon$.*
(2) *If there are open sets $V_1, V_2, \ldots$ such that $A \subset \bigcup_{i=1}^{\infty} V_i$ and $\mu(V_i) < \infty$ for all i, then there is an open set V such that $A \subset V$ and $\mu(V \setminus A) < \varepsilon$.*

Note. The result holds for any Borel measure provided A is a Borel set.

In $\mathbf{R}^n$ it follows immediately that the set C in (1) can be taken to be compact. This holds of course in any σ-compact space X, where every closed set is a countable union of compact sets.

Proof. (1) Replacing μ by the restriction $\mu \llcorner A$ we may assume $\mu(X) < \infty$ by Theorem 1.9 (2). We first verify that all Borel sets A have the required property for all $\varepsilon > 0$ by using the definition of the Borel sets. The first natural attempt would be to show that the family of all A satisfying (1) for all $\varepsilon > 0$ is a σ-algebra, but then we would have problems in showing that it is closed under complementation. Thus we introduce the seemingly smaller family $\mathcal{A}$ of all subsets A of X such that for every $\varepsilon > 0$ there are a closed set C and an open set V such that $C \subset A \subset V$ and $\mu(V \setminus C) < \varepsilon$. It is now rather straightforward to verify that $\mathcal{A}$ is a σ-algebra, which contains the closed sets, and thus also the Borel sets. Hence (1) is established for Borel sets A.

If A is a μ measurable set with $\mu(A) < \infty$ there is a Borel set B such that $A \subset B$ and $\mu(A) = \mu(B)$. Then $\mu(B \setminus A) = 0$ and $B \setminus A$ is contained in a Borel set D with $\mu(D) = 0$. Thus $E = B \setminus D$ is a Borel set with $E \subset A$ and $\mu(A \setminus E) = 0$. Hence knowing (1) for E yields it also for A.

(2) Applying (1) to the sets $V_i \setminus A$ we find closed sets $C_i \subset V_i \setminus A$ such that $\mu(V_i \setminus A \setminus C_i) < \varepsilon/2^i$ for $i = 1, 2, \ldots$. Then $A \subset V = \bigcup_i (V_i \setminus C_i)$, which is an open set with $\mu(V \setminus A) < \varepsilon$. □

1.11. Corollary. *A measure μ on $\mathbf{R}^n$ is a Radon measure if and only if it is locally finite and Borel regular.*

The proof is left as an exercise.

In what follows we shall mainly work with Borel regular measures or Radon measures for convenience. But often they could quite easily be replaced by Borel measures or locally finite Borel measures, for example with the help of Exercise 1.

We shall often encounter measures μ which are carried by a proper subset F of X, that is, $\mu(X \setminus F) = 0$. It is not hard to see that in the case where μ is a Borel measure and X is separable, there exists a unique smallest closed set with this property.

1.12. Definition. *If μ is a Borel measure on a separable metric space X, the support of μ, $\operatorname{spt}\mu$, is the smallest closed set F such that $\mu(X \setminus F) = 0$.*

In other words,

$$\begin{aligned}\operatorname{spt}\mu &= X \setminus \bigcup \{V : V \text{ open}, \mu(V) = 0\} \\ &= X \setminus \{x : \exists\, r > 0 \text{ such that } \mu(B(x,r)) = 0\}.\end{aligned}$$

1.13. Examples. (1) Let f be a non-negative continuous function on $\mathbf{R}^n$. Define a measure μ_f by

$$\mu_f(A) = \int_A f\, d\mathcal{L}^n$$

for $\mathcal{L}^n$ measurable sets A. Then the support of μ_f agrees with that of f:

$$\operatorname{spt}\mu_f = \operatorname{spt} f = \operatorname{Cl}\{x : f(x) \neq 0\},$$

where Cl refers to closure.

(2) Let $Q = \{q_1, q_2, \dots\}$ be an enumeration of the rational numbers, and

$$\mu = \sum_{i=1}^{\infty} 2^{-i}\delta_{q_i},$$

where δ_{q_i} is the Dirac measure at q_i as in 1.6 (2). Then μ is a finite Radon measure on $\mathbf{R}$ with $\operatorname{spt}\mu = \mathbf{R}$. Nevertheless, μ is carried by the countable set Q in the sense that $\mu(\mathbf{R} \setminus Q) = 0$.

Integrals

The integral

$$\int_A f\,d\mu = \int_A f(x)\,d\mu x$$

with respect to a measure μ over a set A of a function f is defined in the usual way, as well as the μ measurability and integrability of f. When the domain of the integration A is the whole space X, we often omit it using the notation

$$\int f\,d\mu = \int_X f\,d\mu.$$

In $\mathbf{R}^n$ we abbreviate the Lebesgue integral

$$\int_A f(x)\,dx = \int_A f(x)\,d\mathcal{L}^n x.$$

The integral $\int f\,d\mu$ is defined for any non-negative μ measurable function on X. Even when $f\colon X \to [0,\infty]$ is not μ measurable we can define the *lower* and *upper integrals* by

$$\int_* f\,d\mu = \sup_{\varphi} \int \varphi\,d\mu \quad \text{and} \quad \int^* f\,d\mu = \inf_{\psi} \int \psi\,d\mu,$$

where φ and ψ run through the μ measurable functions $X \to [0,\infty]$ such that $\varphi \le f \le \psi$.

The μ integrability of $f\colon X \to \overline{\mathbf{R}}$ (in the last two chapters of $f\colon X \to \mathbf{C}$) means that f is μ measurable and $\int |f|\,d\mu < \infty$. As usual, for $1 \le p < \infty$ the space of μ measurable functions $f\colon X \to \overline{\mathbf{R}}$ (or $\mathbf{C}$) with $\int |f|^p\,d\mu < \infty$ is denoted by $L^p(\mu)$, and $L^\infty(\mu)$ is the space of functions which are essentially bounded with respect to μ.

A function $f\colon A \to \overline{\mathbf{R}}$ is a *Borel function* if A is a Borel set and the sets $\{x \in A : f(x) < c\}$ are Borel sets for all $c \in \mathbf{R}$. A mapping $f\colon X \to Y$ between metric spaces X and Y is a Borel mapping if $f^{-1}(U)$ is a Borel set for every open set $U \subset Y$.

We shall mention here only a few of the well-known properties of the integral. The following form of Fubini's theorem will be frequently used.

1.14. Theorem. *Suppose that X and Y are separable metric spaces, and μ and ν are locally finite Borel measures on X and Y, respectively. If f is a non-negative Borel function on $X \times Y$, then*

$$\iint f(x,y)\,d\mu x\,d\nu y = \iint f(x,y)\,d\nu y\,d\mu x.$$

In particular, when f is the characteristic function of a Borel set A,

$$\int \mu\big(\{x : (x,y) \in A\}\big)\,d\nu y = \int \nu\big(\{y : (x,y) \in A\}\big)\,d\mu x.$$

There are many more general forms of Fubini's theorem, see e.g. Hewitt and Stromberg [1], Evans and Gariepy [1, § 1.4], Federer [3, § 2.6] or Jacobs [1, Chapter VI]. To formulate an extension, define the product measure $\mu \times \nu$ by

$$(\mu \times \nu)(C) = \inf \sum_{i=1}^{\infty} \mu(A_i)\,\nu(B_i)$$

where the infimum is taken over all sequences $A_1, A_2, \dots$ of μ measurable sets and $B_1, B_2, \dots$ of ν measurable sets such that

$$C \subset \bigcup_{i=1}^{\infty} A_i \times B_i.$$

Here $0 \cdot \infty = \infty \cdot 0 = 0$. It is easy to see that $\mu \times \nu$ is a measure over $X \times Y$. Moreover, if both μ and ν are either Borel, Borel regular, or Radon measures, $\mu \times \nu$ has the same property. The statement of Theorem 1.14 is valid for all $\mu \times \nu$ measurable functions f which are non-negative or $\mu \times \nu$ integrable (i.e. $\int |f|\,d(\mu \times \nu) < \infty$), and the iterated integrals agree with the $\mu \times \nu$ integral:

$$\int f\,d(\mu \times \nu) = \iint f(x,y)\,d\mu x\,d\nu y.$$

The assumption that X and Y are separable, which of course implies that $X \times Y$ is separable, guarantees that the Borel sets and functions are $\mu \times \nu$ measurable, see e.g. Hewitt and Stromberg [1, Exercise 21.19].

As an application of Fubini's theorem we record the following useful formula.

1.15. Theorem. *Let μ be a Borel measure and f a non-negative Borel function on a separable metric space X. Then*

$$\int f\,d\mu = \int_0^\infty \mu(\{x \in X : f(x) \geq t\})\,dt.$$

Proof. Let $A = \{(x,t) : f(x) \geq t\}$. Then

$$\int_0^\infty \mu(\{x \in X : f(x) \geq t\})\,dt = \int_0^\infty \mu(\{x : (x,t) \in A\})\,dt$$
$$= \int \mathcal{L}^1(\{t \in [0,\infty) : (x,t) \in A\})\,d\mu x = \int \mathcal{L}^1([0, f(x)])\,d\mu x$$
$$= \int f(x)\,d\mu x.$$

□

Another way to look at the Radon measures and integrals with respect to them is to consider them as linear functionals on $C_0(X)$, the space of compactly supported continuous real-valued functions on X. That is, if μ is a Radon measure on X, we can associate to it the linear functional

$$L\colon C_0(X) \to \mathbf{R}, \quad Lf = \int f\,d\mu.$$

This is obviously *positive* in the sense that

$$Lf \geq 0 \quad \text{for } f \geq 0.$$

In the case where X is locally compact the converse also holds, see e.g. Rudin [1, 2.14].

1.16. Riesz representation theorem. *Let X be a locally compact metric space and $L\colon C_0(X) \to \mathbf{R}$ a positive linear functional. Then there is a unique Radon measure μ such that*

$$Lf = \int f\,d\mu \quad \text{for } f \in C_0(X).$$

Image measures

We can map measures from one metric space X to another, Y.

1.17. Definition. *The image of a measure μ under a mapping $f\colon X \to Y$ is defined by*

$$f_\sharp\mu(A) = \mu(f^{-1}A) \quad \text{for } A \subset Y.$$

It is apparent that $f_\sharp\mu$ is a measure on Y. It is also immediate that A is $f_\sharp\mu$ measurable whenever $f^{-1}(A)$ is μ measurable. Hence if μ is a Borel measure and f a Borel function, $f_\sharp\mu$ is a Borel measure. The following simple criterion on the Radonness of $f_\sharp\mu$ will suffice for us. For more general results, see e.g. Federer [3, 2.2.17] or Schwartz [1, Part I, § I.5].

1.18. Theorem. *Let X and Y be separable metric spaces. If $f\colon X \to Y$ is continuous and μ is a Radon measure on X with compact support, then $f_\sharp\mu$ is a Radon measure. Moreover,* $\operatorname{spt} f_\sharp\mu = f(\operatorname{spt}\mu)$.

Proof. Replacing X by the subspace $\operatorname{spt}\mu$ we may assume X is compact. Statement (i) of Definition 1.5(4) is trivial, as μ, and hence also $f_\sharp\mu$, are finite measures. We leave (ii) as an exercise and prove only (iii).

Let $A \subset Y$ and $\varepsilon > 0$. Since μ is a Radon measure there is an open set $U \subset X$ such that $f^{-1}A \subset U$ and $\mu(U) \le \mu(f^{-1}A) + \varepsilon$. Set $V = Y \setminus f(X \setminus U)$. Then V is open, as X is compact, $A \subset V$ and

$$\begin{aligned} f_\sharp\mu(V) &= \mu(f^{-1}(Y \setminus f(X \setminus U))) \\ &= \mu(X \setminus f^{-1}(f(X \setminus U))) \le \mu(U) \\ &\le \mu(f^{-1}A) + \varepsilon = f_\sharp\mu(A) + \varepsilon. \end{aligned}$$

This yields (iii). We leave the last statement on supports also as an exercise. □

The following theorem can be proven via a rather straightforward approximation by simple functions. It can also be easily deduced from Theorem 1.15.

1.19. Theorem. *Suppose $f\colon X \to Y$ is a Borel mapping, μ is a Borel measure on X, and g is a non-negative Borel function on Y. Then*

$$\int g \, df_\sharp\mu = \int (g \circ f) \, d\mu.$$

When Y is locally compact, all this could also be done in the reverse order: letting

$$Lg = \int (g \circ f) \, d\mu \quad \text{for } g \in C_0(Y)$$

we obtain a linear functional on $C_0(Y)$ which by the Riesz representation theorem 1.16 corresponds to a Radon measure $f_\sharp\mu$.

It is clear that pulling back measures is not nearly as natural as pushing them forward: the formula $\mu(A) = \nu(fA)$ does not usually define a Borel measure even for very nice measures ν if f fails to be injective. Still it is often possible to find such pull-backs abstractly. The following proof can be found in Schwartz [1, Part I, § I.5] and in Fuglede [1].

1.20. Theorem. *Let X and Y be compact metric spaces and $f\colon X \to Y$ a continuous surjection. For any Radon measure ν on Y there exists a Radon measure μ on X such that $f_\sharp\mu = \nu$.*

Proof. We define a functional $p\colon C_0(X) \to \mathbf{R}$ by

$$p(\varphi) = \nu(Y)\max\{\varphi(x) : x \in X\}.$$

Clearly p is a seminorm, that is, $p(\varphi + \psi) \leq p(\varphi) + p(\psi)$ and $p(\lambda\varphi) = \lambda p(\varphi)$ for $\varphi, \psi \in C_0(X)$ and $0 \leq \lambda < \infty$. Since f is surjective we can define a linear form λ on the vector subspace $\{\psi \circ f : \psi \in C_0(Y)\}$ by

$$\lambda(\psi \circ f) = \int \psi \, d\nu.$$

It satisfies

$$\begin{aligned}\lambda(\psi \circ f) &\leq \nu(Y)\max\{\psi(y) : y \in Y\}\\ &= \nu(Y)\max\{(\psi \circ f)(x) : x \in X\} = p(\psi \circ f).\end{aligned}$$

By the Hahn–Banach theorem, see e.g. Rudin [2, Theorem 3.2], λ extends to a linear form μ on $C_0(X)$ satisfying

$$\mu(\varphi) \leq p(\varphi) \quad \text{for } \varphi \in C_0(X).$$

In particular, $\mu(\varphi) \leq 0$ if $\varphi \leq 0$, whence $\mu(\varphi) = -\mu(-\varphi) \geq 0$, if $\varphi \geq 0$. By the Riesz representation theorem 1.16 μ can be identified with a Radon measure. As μ extends λ, we have, by Theorem 1.19,

$$\int \psi \, d\nu = \int (\psi \circ f) \, d\mu = \int \psi \, df_\sharp\mu \quad \text{for } \psi \in C_0(Y).$$

Thus $f_\sharp\mu = \nu$ by the uniqueness part of the Riesz representation theorem 1.16. □

Weak convergence

Next we consider a convergence of measures.

1.21. Definition. *Let $\mu, \mu_1, \mu_2, \dots$ be Radon measures on a metric space X. We say that the sequence (μ_i) converges weakly to μ,*

$$\mu_i \xrightarrow{\mathrm{w}} \mu,$$

if

$$\lim_{i\to\infty} \int \varphi \, d\mu_i = \int \varphi \, d\mu \quad \text{for all } \varphi \in C_0(X).$$

1.22. Examples.

(1) In $\mathbf{R}$, $\delta_i \xrightarrow{\mathrm{w}} 0$ as $i \to \infty$.
(2) Let

$$\mu_k = \frac{1}{k} \sum_{i=1}^{k} \delta_{i/k}.$$

Then $\mu_k \xrightarrow{\mathrm{w}} \mathcal{L}^1 \llcorner [0,1]$.

The weak convergence is useful because a very general compactness theorem holds. We prove it only for $\mathbf{R}^n$.

1.23. Theorem. *If $\mu_1, \mu_2, \dots$ are Radon measures on $\mathbf{R}^n$ with*

$$\sup\{\mu_i(K) : i = 1, 2, \dots\} < \infty$$

for all compact sets $K \subset \mathbf{R}^n$, then there is a weakly convergent subsequence of (μ_i).

Proof. The space $C_0(\mathbf{R}^n)$ is separable under the norm

$$\|\varphi\| = \max\left\{|\varphi(x)| : x \in \mathbf{R}^n\right\},$$

whence it has a countable dense subset D. For example, choosing functions $\varphi_i \in C_0(\mathbf{R}^n)$, $i = 1, 2, \dots$, with $\varphi_i = 1$ on $B(i)$, one can by the Weierstrass approximation theorem take for D the set of all products $\varphi_i P$ where $i = 1, 2, \dots$ and P runs through polynomials with rational coefficients. For each $\varphi \in D$ the bounded sequence $(\int \varphi \, d\mu_i)$ of real numbers has a convergent sub-sequence. Using the diagonal method we can thus extract a sub-sequence (μ_{i_k}) such that the limit

$$L\varphi = \lim_{k\to\infty} \int \varphi \, d\mu_{i_k}$$

exists and is finite for all $\varphi \in D$. The denseness of D then implies that this actually holds for all $\varphi \in C_0(\mathbf{R}^n)$, and the Riesz representation theorem 1.16 gives the limit measure. □

As Example 1.22 shows $\mu_i \xrightarrow{w} \mu$ need not imply that $\mu_i(A) \to \mu(A)$ even when $A = \mathbf{R}^n$, see also Exercise 9. However, the following semi-continuity properties hold.

1.24. Theorem. *Let $\mu_1, \mu_2, \ldots$ be Radon measures on a locally compact metric space. If $\mu_i \xrightarrow{w} \mu$, $K \subset X$ is compact and $G \subset X$ is open, then*

(1) $$\mu(K) \geq \limsup_{i\to\infty} \mu_i(K),$$

(2) $$\mu(G) \leq \liminf_{i\to\infty} \mu_i(G).$$

Proof. (1) Let $\varepsilon > 0$. By property (4) (iii) of Definition 1.5 there is an open set V such that $K \subset V$ and $\mu(V) \leq \mu(K) + \varepsilon$. By Urysohn's lemma, see e.g. Rudin [1, 2.12], there is $\varphi \in C_0(X)$ such that $0 \leq \varphi \leq 1$, $\varphi = 1$ on K and $\operatorname{spt} \varphi \subset V$. Thus

$$\begin{aligned}\mu(K) &\geq \mu(V) - \varepsilon \geq \int \varphi \, d\mu - \varepsilon \\ &= \lim_{i\to\infty} \int \varphi \, d\mu_i - \varepsilon \geq \limsup_{i\to\infty} \mu_i(K) - \varepsilon,\end{aligned}$$

and (1) follows.

(2) is proven similarly through approximation of G with compact sets from inside. □

Approximate identities

We shall now show that arbitrary Radon measures in $\mathbf{R}^n$ can be approximated weakly by smooth functions, that is, by measures of the form $A \mapsto \int_A g \, d\mathcal{L}^n$ where $g \in C^\infty(\mathbf{R}^n)$, the space of infinitely differentiable real-valued functions on $\mathbf{R}^n$. First we define convolutions.

1.25. Definition. *Let f and g be real-valued functions on $\mathbf{R}^n$ and μ a Radon measure on $\mathbf{R}^n$. The convolutions $f * g$ of f and g, and $f * \mu$ of f and μ, are defined by*

$$f * g(x) = \int f(x-y)\, g(y)\, dy,$$
$$f * \mu(x) = \int f(x-y)\, d\mu y,$$

provided the integral exists.

We now consider an *approximate identity* $\{\psi_\varepsilon\}_{\varepsilon>0}$. By this we mean that each ψ_ε is a non-negative continuous function on $\mathbf{R}^n$ such that

$$\operatorname{spt}\psi_\varepsilon \subset B(\varepsilon) \quad \text{and} \quad \int \psi_\varepsilon\, d\mathcal{L}^n = 1.$$

Any continuous function $\psi\colon \mathbf{R}^n \to [0,\infty)$ with $\operatorname{spt}\psi \subset B(1)$ and $\int \psi\, d\mathcal{L}^n = 1$ obviously gives such an approximate identity by

$$\psi_\varepsilon(x) = \varepsilon^{-n}\psi(x/\varepsilon).$$

In particular we may take

$$\begin{aligned} \psi_\varepsilon(x) &= c(\varepsilon)\, e^{-1/(\varepsilon^2-|x|^2)} && \text{for } |x| < \varepsilon, \\ \psi_\varepsilon(x) &= 0 && \text{for } |x| \geq \varepsilon, \end{aligned}$$

where $c(\varepsilon)$ is determined by $\int \psi_\varepsilon\, d\mathcal{L}^n = 1$, to get an approximate identity consisting of C^∞ functions. It is shown in many text-books that for any such approximate identity consisting of C^∞ functions the functions $\psi_\varepsilon * f$, where $f \in L^p(\mathbf{R}^n)$, are also C^∞ and they converge to f in L^p. We now study $\psi_\varepsilon * \mu$ in the same spirit.

1.26. Theorem. *Let $\{\psi_\varepsilon\}_{\varepsilon>0}$ be an approximate identity and μ a Radon measure on $\mathbf{R}^n$. Then the functions $\psi_\varepsilon * \mu$ are infinitely differentiable and they converge weakly to μ as $\varepsilon \downarrow 0$, that is,*

$$\lim_{\varepsilon\downarrow 0} \int \varphi(\psi_\varepsilon * \mu)\, d\mathcal{L}^n = \int \varphi\, d\mu$$

for all $\varphi \in C_0(\mathbf{R}^n)$. If $\mu(\mathbf{R}^n) < \infty$, this holds for all uniformly continuous bounded functions $\varphi\colon \mathbf{R}^n \to \mathbf{R}$.

Proof. By studying the difference quotients and using induction one can verify in a straightforward manner that for all $i_j \in \{1,\ldots,n\}$, $j = 1,\ldots,k$,

$$\partial_{i_1}\ldots\partial_{i_k}(\psi_\varepsilon * \mu) = (\partial_{i_1}\ldots\partial_{i_k}\psi_\varepsilon) * \mu,$$

where ∂_i means the partial derivative with respect to the i-th coordinate. It follows that $\psi_\varepsilon * \mu$ has partial derivatives of all orders.

To prove the second statement we use Fubini's theorem, change of variable and the facts that $\operatorname{spt}\psi_\varepsilon \subset B(\varepsilon)$ and $\int \psi_\varepsilon \, d\mathcal{L}^n = 1$ to compute

$$\begin{aligned}
&\int \varphi(\psi_\varepsilon * \mu)\, d\mathcal{L}^n - \int \varphi \, d\mu \\
&= \int \varphi(x) \int \psi_\varepsilon(x-y)\, d\mu y \, dx - \int \varphi(y) \int \psi_\varepsilon(x)\, dx \, d\mu y \\
&= \int \left[\int \varphi(x)\, \psi_\varepsilon(x-y)\, dx - \int \varphi(y)\, \psi_\varepsilon(x)\, dx \right] d\mu y \\
&= \iint_{B(\varepsilon)} [\varphi(x+y) - \varphi(y)]\, \psi_\varepsilon(x)\, dx \, d\mu y.
\end{aligned}$$

Since φ is uniformly continuous with compact support and $\int \psi_\varepsilon \, d\mathcal{L}^n = 1$, this goes to zero as $\varepsilon \downarrow 0$. The last statement follows also by the above proof. □

We finish this chapter with some remarks on lower semicontinuous functions. We shall need this concept only for non-negative functions. One way to define them is to say that a non-negative function g on $\mathbf{R}^n$ is *lower semicontinuous* if there are non-negative functions $\varphi_i \in C_0(\mathbf{R}^n)$, $i = 1, 2, \ldots$, such that $\varphi_1 \le \varphi_2 \le \ldots$ and $g = \lim_{i\to\infty} \varphi_i$. An equivalent definition is that the sets $\{x : g(x) > c\}$ are open for all $c \in \mathbf{R}$. Examples are characteristic functions of open sets and $x \mapsto |x|^p$, $p \in \mathbf{R}$ (with value ∞ at 0 if $p < 0$).

The following simple lemma will be needed in Chapter 12.

1.27. Lemma. *Let $\{\psi_\varepsilon\}_{\varepsilon>0}$ be an approximate identity, μ a Radon measure on $\mathbf{R}^n$, and g a non-negative lower semicontinuous function on $\mathbf{R}^n$. Then*

$$\int (g * \mu)\, d\mu \le \liminf_{\varepsilon \downarrow 0} \int [(g * \psi_\varepsilon) * \mu]\, d\mu.$$

Proof. Approximating μ by the restrictions $\mu \llcorner B(k)$, $k = 1, 2, \ldots$, we may assume that μ has compact support. For $\varphi \in C_0(\mathbf{R}^n)$, $\varphi * \psi_\varepsilon \to \varphi$ uniformly as $\varepsilon \downarrow 0$, whence

$$\int (\varphi * \mu)\, d\mu = \lim_{\varepsilon \downarrow 0} \int [(\varphi * \psi_\varepsilon) * \mu]\, d\mu.$$

Applying this, the definition of lower semicontinuity and the monotone convergence theorem, we get the required inequality. □

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Exercises.

1. Show that ν^* defined by (1.2) is a measure agreeing with ν on $\mathcal{A}$, and, moreover, that ν^* is Borel regular if $\mathcal{A}$ is contained in the family of Borel sets.
2. Show that if μ is a measure, $A \subset B$, B is μ measurable and $\mu(A) = \mu(B) < \infty$, then $\mu(A \cap C) = \mu(B \cap C)$ for all μ measurable sets C.
3. Verify the statements in Examples 1.6 (2) and (3) concerning Dirac and counting measures. What are the measurable sets with respect to them?
4. Prove that a Borel measure μ satisfies the condition of Theorem 1.7.
5. Prove Corollary 1.11.
6. Show that $f_\sharp\mu$ in Theorem 1.18 satisfies condition (ii) of Definition 1.5 (4), and that $\operatorname{spt} f_\sharp\mu = f(\operatorname{spt}\mu)$.
7. Let μ be a finite Borel measure on $\mathbf{R}^n$ with compact support. Show that there exists $b \in \mathbf{R}^n$, the centre of mass of μ, such that
$$b \cdot v = \left(\int x \cdot v \, d\mu x \right) \Big/ \mu(\mathbf{R}^n) \quad \text{for } v \in \mathbf{R}^n.$$
8. Let $\{\psi_\varepsilon\}_{\varepsilon>0}$ be an approximate identity in $\mathbf{R}^n$ and $\varphi \in C_0(\mathbf{R}^n)$. Show that $\psi_\varepsilon * \varphi \to \varphi$ uniformly as $\varepsilon \downarrow 0$.
9. Let $\mu_1, \mu_2, \dots$ and μ be Radon measures on $\mathbf{R}^n$ with $\mu_i \xrightarrow{\text{w}} \mu$. Show that if A is a bounded subset of $\mathbf{R}^n$ and $\mu(\partial A) = 0$, then $\lim_{i\to\infty} \mu_i(A) = \mu(A)$.